METRIC

MIL-STD-1399
SECTION 407
14 July 2022

SUPERSEDING
DOD-STD-1399(NAVY)
SECTION 070, PART 1
26 February 1979

DEPARTMENT OF DEFENSE INTERFACE STANDARD

SECTION 407

DC MAGNETIC FIELD ENVIRONMENT

AMSC N/A FSC 1990

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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. This section defines the standard interface requirements for and the constraints on the design of shipboard equipment whose performance may be degraded when subjected to the shipboard direct current (DC) magnetic field environment.

3. Equipment installed on board a naval ship can be expected to encounter a magnetic field environment that would not be encountered in other circumstances. This field is generated by efforts to counteract the normal magnetic signature of a metallic ship (i.e., by degaussing) and may have a degrading effect on the performance of susceptible equipment.

4. The technical content first delineates the characteristics in terms of the magnetic field strength and distribution, then establishes the constraints on equipment design and installation that are necessary to achieve shipboard compatibility with these characteristics.

5. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This section establishes interface requirements for shipboard equipment whose performance may be degraded when subjected to the ship's DC magnetic field to ensure compatibility between such equipment and the magnetic field environment. The policies and procedures established by MIL-STD-1399 are mandatory. This section and the basic standard are to be viewed as an integral single document.

1.2 Interface. Basic characteristics and constraint categories concerned with this interface are shown symbolically on figure 1 (see MIL-STD-1399). The interface characteristics and constraints pertinent to this section are described in 5.2 and 5.3.

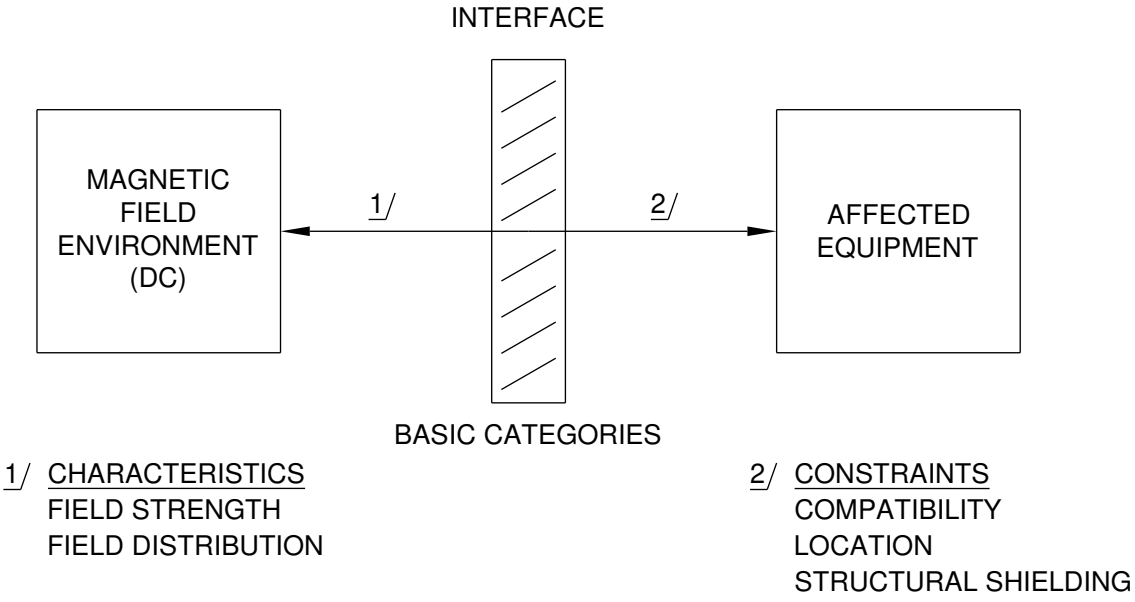


FIGURE 1. Interface.

1.3 Applicability. The criteria of this section apply to all shipboard equipment — whether permanently installed, temporarily installed, or transported — whose performance may be degraded when subjected to the magnetic fields defined herein. Such magnetic fields will normally be experienced in surface ships and submarines that have a degaussing installation, and in all ships that are or may be subjected to magnetic treatment (see 3.1.2 and 5.3.3).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

MIL-STD-464 - Electromagnetic Environmental Effects Requirements for Systems

MIL-STD-1399 - Interface Standard for Shipboard Systems

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-419 - Grounding, Bonding, and Shielding for Electronic Equipments and Facilities

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9407-AB-HBK-010 - Handbook of Shipboard Electromagnetic Shielding Practices

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

3. DEFINITIONS

3.1 Degaussing. Degaussing is the science of reducing a ship's inherent DC magnetic signature so that the possibility of detection by magnetic mines and other magnetic influence detection devices is minimized.

3.1.1 Degaussing system. A degaussing system is a shipboard system designed to reduce the ship's DC magnetic signature. Typical degaussing systems consist of control equipment, power supply equipment, and one or more coils of electrical cable installed at specific locations aboard a ship that vary the magnetic field in both magnitude and polarity.

3.1.2 Magnetic treatment. Degaussing by magnetic treatment is a method of altering or reducing the permanent magnetic field of a ship by temporarily rigging coils of electrical cable externally around and about the ship, including the superstructure where necessary, and energizing the coils with a sequence of DC pulses of predetermined polarities and magnitudes. "Deperming" and "flashing" are types of magnetic treatment.

3.1.2.1 Deperming. Deperming is the process whereby a ship's permanent longitudinal and athwartship magnetism is ideally removed and its vertical permanent magnetism is stabilized at a known level by exposing the platform to large magnetic fields of alternating polarities and decreasing magnitude.

3.1.2.2 Flashing. Flashing is a deperming treatment used on ships and submarines with no shipboard degaussing coils. In this treatment, the vessel's permanent vertical magnetism is changed in such a way that the total magnetization will be at a minimum in its intended area of operation.

3.2 Equipment. Equipment as referred to in this section comprises those shipboard components, equipments, and systems whose performance may be degraded when subjected to a magnetic field environment as specified herein. This equipment may be permanently installed, temporarily installed, or transported aboard the ship.

3.3 Magnetic field strength. Magnetic field strength is expressed in amperes per meter (A/m). The unit of field strength in the centimeter-gram-second (CGS) system is the oersted (Oe), which corresponds to one line of force per square centimeter (cm^2) of area perpendicular to that line of force. One Oe is approximately equal to 80 A/m.

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3.4 Ship's DC magnetic field. A ship's DC magnetic field is the region in and around the ship in which magnetic forces are observable. The magnitude of this field varies with both time and location.

4. GENERAL REQUIREMENTS

4.1 Operational DC magnetic environment. Equipment used aboard ships shall not be degraded when exposed to its operational DC magnetic environment when tested in accordance with MIL-STD-464 and as specified herein.

4.2 Requirements. The specific interface requirements and constraints established herein are mandatory and shall be adhered to by Systems Commands (SYSCOM), Project Managers, contractors, and all others engaged in any aspect of shipboard design to which these requirements and constraints apply, including equipment design, production, and installation (see MIL-STD-1399).

5. DETAILED REQUIREMENTS

5.1 General considerations. The shipboard magnetic field environment is not uniform and will vary both in location and time as a result of varying electromagnetic sources. If this field is of sufficient magnitude in any given instance, it may degrade the performance of equipment installed in or on the ship. Detailed knowledge of the exact installation location, as well as a forecast of future changes to this location, would be involved if a determination of equipment suitability were to be made on other than general environmental expectations. Equipment shall be so designed and built that it will not be adversely affected when the ship undergoes magnetic treatment or operates with degaussing coils energized.

5.2 Interface characteristics. The interface characteristics of the ship's DC magnetic field environment shall be as described in 5.2.1, 5.2.2, and 5.2.3 and may be expected to be present where the equipment interfaces with the shipboard DC magnetic field environment.

5.2.1 Field strength. The steady state magnetic field strength that equipment shall be expected to withstand is 1,600 A/m. No particular orientation of the direction of the magnetic field shall be assumed.

5.2.2 Field rate of change. The expected magnetic field rate of change shall be no less than 1,600 A/m per second. Note that exceeding this rate by a factor of 10 may overstimulate the system under test and potentially cause damage.

5.2.3 Field distribution. The magnetic field strength delineated in 5.2.1 is not uniform throughout the ship and is most likely to occur near degaussing cables. During magnetic treatment of the ship (see 3.1.2), these field values may occur over the entire ship. Other factors, such as proximity to degaussing cables or certain motors, generators, or other electrical devices, may also affect the field distribution (see 5.3).

5.3 Interface constraints. Interface characteristics of the ship's DC magnetic field environment impose constraints on the design and location of shipboard equipment. These constraints shall be as described in 5.3.1 through 5.3.3.

5.3.1 Compatibility. The design of equipment shall be compatible with the interface characteristics given in 5.2.

5.3.2 Location considerations. Since the magnetic field is not uniform, consideration shall be given to its variability in determining the location of shipboard equipment. The situations that shall be considered are discussed in 5.3.2.1 through 5.3.2.4.

5.3.2.1 Proximity to degaussing cables. Field strength in excess of 1,600 A/m may be found at distances within 0.3 meter from any degaussing cable. Equipment that must be located within 0.3 meter of a degaussing cable shall be evaluated to determine the appropriate field level required for performance testing (see 5.5).

5.3.2.2 Local magnetic fields. Local magnetic field strengths in excess of 1,600 A/m can be generated by various individual sources, such as:

a. Electrical power cables

b. Generators

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c. Motors

d. Welding circuits

e. Electrical power switchboards and control equipment

f. Degaussing system installations

Location of equipment in the proximity of those sources should be judiciously considered.

5.3.2.3 Structural discontinuities. Bulkheads or other steel structural members (including equipment) may act as a return path for the magnetic flux lines, resulting in a localized concentration of the magnetic field. If there is a discontinuity (gaps, openings, etc.) in this return path, increased field values may be experienced in the vicinity of the discontinuity. When equipment is to be installed in the near vicinity of such a discontinuity, the possible degrading effect of the enhanced magnetic field environment shall be taken into account.

5.3.2.4 Structural shielding. Steel decks and bulkheads between degaussing cables or other sources of magnetic fields and the equipment may provide some degree of shielding and thereby reduce the magnetic field that would otherwise exist. However, taking advantage of this shielding effect would require precise knowledge of the location of the degaussing cables or other sources in relation to the location of the equipment and will require rigid control over location changes. This shielding effect shall not be utilized to exempt any equipment from the requirements of this section. However, in cases where it has been established that it is not technically feasible for equipment to meet the requirements of this section, it may be appropriate to submit a request for deviation utilizing structural shielding as a justification (see 5.5).

5.3.2.5 Shielding enclosures. Equipment that is sensitive to magnetic fields may require a magnetic shielding enclosure composed of materials having a high relative magnetic permeability (mu), such as steel, mu-metal, permalloy, or other high-mu alloys and composites. Aluminum and copper are ineffective for DC or low-frequency magnetic shielding and shall not be used for this purpose. The use of MIL-HDBK-419 or S9407-AB-HBK-010 for guidance on shielding practices is recommended but is not required.

5.3.3 Special circumstances. Surface ships and submarines that do not have degaussing installations are subject to magnetic treatment at periodic intervals. This treatment is accomplished at dockside and does not require equipment to be operating during such treatment. The equipment shall operate in accordance with applicable component design specifications in the presence of a residual field of 400 A/m after exposure to the magnetic treatment. In certain circumstances, it may not be practical or cost-effective to design specialized equipment to operate in the maximum expected field. In such cases, the need to deviate from the requirement of 5.3.1 shall be fully justified (see 5.5).

5.4 Performance testing.

5.4.1 Requirement. When it is considered necessary to determine if equipment (see 3.2) will meet the criteria of this section, performance tests shall be conducted. Such shipboard equipment shall be tested by exposing it to a steady state magnetic field having a magnitude of at least 1,600 A/m over the entire volume of the equipment under test. The minimum exposure to the steady state field shall be 5 minutes. The magnetic field shall be increased from 0 A/m to the maximum magnitude at a rate no less than 1,600 A/m per second at the start of the 5-minute period. The magnetic field shall be decreased from the maximum magnitude to 0 A/m at a rate no less than 1,600 A/m per second at the end of the 5-minute period. Exceeding the minimum rate of change by a factor of 10 has potential to cause damage to the equipment. No particular orientation of the direction of the magnetic field shall be assumed. Equipment shall be tested while operating and the equipment performance evaluated in the presence of the test field. Requirements for testing individual equipment shall be specified in the individual equipment specification.

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5.4.2 Test procedure. A coil (typically square or circular) of electrical conductors large enough to surround the equipment shall be provided (see Appendix A). In lieu of a single coil, Helmholtz or Rubens coils may be used to satisfy the coil requirements. (Note: The criteria contained in Appendix A are not applicable to the Helmholtz or Rubens coils.) The coil shall contain enough ampere-turns to develop a magnetic field of at least 1,600 A/m over the entire volume of the equipment under test when energized by a power source. The coil shall be calibrated for the maximum desired steady state magnetic field in air, without the equipment in the field. While this is typically 1,600 A/m, a higher steady state magnetic field may be needed to meet the criteria in this standard. The power source shall be capable of providing a variable (single polarity) DC. The equipment under test shall be positioned at the center of the coil and the test shall be performed as follows:

a. Increase the magnetic field strength from 0 A/m to at least 1,600 A/m at a rate of at least 1,600 A/m per second. Maintain the field strength above 1,600 A/m for 5 minutes, or longer if necessary, to obtain sufficient operational data from the equipment to verify correct operation. Decrease the magnetic field strength from 1,600 A/m to 0 A/m at a rate of at least 1,600 A/m per second. As noted earlier, do not exceed the minimum rate of change by more than a factor of 10; equipment damage may result.

b. Reverse the connections of the coil to the power source and repeat the test procedure stated in (a).

c. Rotate the coil 90 degrees such that the plane of the coil is perpendicular to the plane of the coil as stated in (a) and repeat the test procedure as stated in (a) and (b). If more convenient, the equipment rather than the coil may be rotated.

d. Rotate the coil 90 degrees such that the plane of the coil is perpendicular to the plane of the coil as stated in both (a) and (c) and repeat the test procedure as stated in (a) and (b). If more convenient, the equipment rather than the coil may be rotated.

5.5 Tailoring of requirements. Application-specific environmental criteria may be derived from operational and engineering analyses on equipment or subsystems being procured for use in specific systems or platforms. When analyses reveal that the requirements in this standard are not appropriate for that procurement (see 6.2), the requirements shall be tailored prior to the start of the test program. The test procedures contained in this document are generic test methods and shall be adapted as necessary for each application while maintaining the intent of the test. All such adaptations shall be approved by the procuring activity. When specified (see 6.2), the adapted test procedures shall be documented in the Electromagnetic Interference Test Procedures (EMITP) in accordance with MIL-STD-461.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard is intended for use in the acquisition cycle of equipment and subsystems to specify the DC magnetic field susceptibility requirements for the control of electromagnetic interference (EMI).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Whether tailoring requirements to be application-specific is required (see 5.5).

c. Whether adapted test procedures are required to be documented in the EMITP in accordance with MIL-STD-461 (see 5.5).

6.3 Subject term (key word) listing.

Degaussing

Deperming

Flashing

Magnetic field, low frequency

Magnetic treatment

Susceptibility, magnetic field

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6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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COIL DESIGN PROCEDURE AND CRITERIA

A.1 SCOPE

A.1.1 Scope. This appendix is not a mandatory part of the standard. The information contained herein is intended for guidance only. It delineates the mathematical approach for determining the minimum diameter of a coil to generate a magnetic field of 1,600 A/m at the center of the coil in air (i.e., without the equipment in the field) and a field of some larger predetermined value at the surface of the equipment nearest the coil. Detail design of a coil is beyond the scope of this appendix since actual coil design parameters are dependent on equipment size and available power, voltage, current, materials, space, etc.

A.2 EQUATIONS

A.2.1 Ampere's formula. In a nonmagnetic medium, the magnetic field intensity dH produced at a point by an element of a conductor dℓ in meters (m) through which there is a current of I amperes is:

EQUATION 1

$$dH = \frac{I \sin \theta \, d\ell}{4\pi r^2} \qquad A/m$$

Where r is the distance, in meters, between the element dℓ and the point, and Θ is the angle between the directions of dℓ and r (see figure A-1).

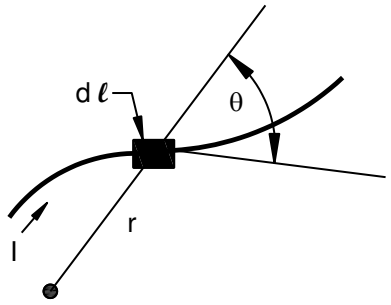


FIGURE A-1. Ampere's formula.

A.2.2 Coil design. The magnetic field intensity, in air, produced at a point by a closed circular coil can be derived using ampere's formula. The derivation for a point inside the coil and in the plane of the coil is as follows (see figure A-2):

a. A – point at center of coil.
b. B – tangent point on circumference.
c. C – point in plane of coil and inside coil.
d. R – mean radius of coil.

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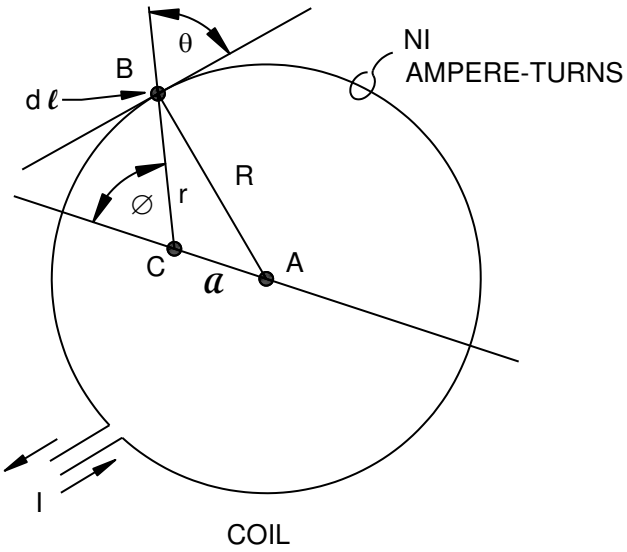


FIGURE A-2. Coil design.

EQUATION 2

Solving for r as a function of ø (from the law of cosines for triangle ABC):

$$R^2 = r^2 + a^2 - 2ar\cos(180 - ø)$$

$$R^2 = r^2 + a^2 + 2ar\cos ø$$

$$0 = r^2 + (2a\cos ø)r + (a^2 - R^2)$$

$$r = \frac{-2a\cos ø \pm \sqrt{4a^2\cos^2 ø - 4(a^2 - R^2)}}{2}$$

$$r = -a\cos ø + \sqrt{R^2 - a^2\sin^2 ø}$$

EQUATION 3

Solving for Θ as a function of ø (from the law of sines for triangle ABC):

$$\frac{\sin(180 - ø)}{R} = \frac{\sin(90 - Θ)}{a}$$

$$\frac{\sin ø}{R} = \frac{\cos Θ}{a}$$

$$\cos Θ = \frac{a}{R}\sin ø$$

$$\sqrt{1 - \sin^2 Θ} = \frac{a}{R}\sin ø$$

$$\sin^2 Θ = 1 - \frac{a^2}{R^2}\sin^2 ø$$

$$\sin Θ = \frac{1}{R}\sqrt{R^2 - a^2\sin^2 ø}$$

EQUATION 4

And $dℓ \approx r\,dø$ (see figure for validity of approximation)

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Substituting equations 3 and 4 into equation 1:

$$dH = \frac{I}{4\pi r^2}\frac{1}{R}\sqrt{R^2 - a^2 \sin^2\phi}\; r\, d\phi$$

$$dH = \frac{I}{4\pi R}\frac{\sqrt{R^2 - a^2 \sin^2\phi}}{r}\, d\phi$$

Now substituting r from equation 2 gives:

$$dH = \frac{I}{4\pi R}\frac{\sqrt{R^2 - a^2 \sin^2\phi}}{\sqrt{R^2 - a^2 \sin^2\phi} - a\cos\phi}\, d\phi$$

$$H = 2\int_0^{\pi} \frac{I}{4\pi R}\left(1 + \frac{a\cos\phi}{\sqrt{R^2 - a^2 \sin^2\phi} - a\cos\phi}\right) d\phi$$

Rationalizing the denominator gives:

$$H = \frac{2I}{4\pi R}\int_0^{\pi}\left[1 + \frac{a\cos\phi\sqrt{R^2 - a^2 \sin^2\phi} + a^2\cos^2\phi}{R^2 - a^2\sin^2\phi - a^2\cos^2\phi}\right] d\phi$$

$$H = \frac{2I}{4\pi R}\left[\int_0^{\pi} d\phi + \frac{1}{R^2 - a^2}\left(\int_0^{\pi}\sqrt{R^2 - a^2\sin^2\phi}\; a\cos\phi\, d\phi + a^2\int_0^{\pi}\cos^2\phi\, d\phi\right)\right]$$

Integrating the second term using the substitution:
$a \sin\phi = u, a\cos\phi\, d\phi = du,$ with the limits from 0 to 0, the term goes to 0.

$$H = \frac{2I}{4\pi R}\left[\phi + \frac{a^2}{R^2 - a^2}\left(\frac{1}{2}\phi + \frac{1}{4}\sin 2\phi\right)\right]_0^{\pi}$$

$$= \frac{2I}{4\pi R}\left[\phi + \frac{a^2}{2(R^2 - a^2)}\left(\phi + \frac{1}{2}\sin 2\phi\right)\right]_0^{\pi}$$

$$= \frac{2I}{4\pi R}\left[\pi + \frac{a^2}{2(R^2 - a^2)}\,\pi\right]$$

$$= \frac{2\pi I}{4\pi R}\left[\frac{2R^2 - 2a^2 + a^2}{2(R^2 - a^2)}\right]$$

$$H = \frac{\pi I}{4\pi R}\left[\frac{2R^2 - a^2}{R^2 - a^2}\right]$$

EQUATION 5

For a coil with N turns:

$$H = \frac{\pi NI}{4\pi R}\left(\frac{2R^2 - a^2}{R^2 - a^2}\right) = \frac{NI}{4R}\left(\frac{2R^2 - a^2}{R^2 - a^2}\right)$$

EQUATION 6

The magnetic field intensity at the center of the coil (a = 0) is:

$$H = \frac{2\pi NI}{4\pi R} = \frac{NI}{2R}\ \text{A/m}$$

EQUATION 7

A.2.3 Maximum field. The point of maximum field is the point on the surface of the equipment nearest the coil. Equipment dimensions are ℓ, w, and h, where $\ell > w > h$ (see figure A-3). Let "a" equal the longest distance from the center of the equipment to a point on the surface of the equipment in the plane of the coil:

$$a = \sqrt{(\ell/2)^2 + (w/2)^2}$$

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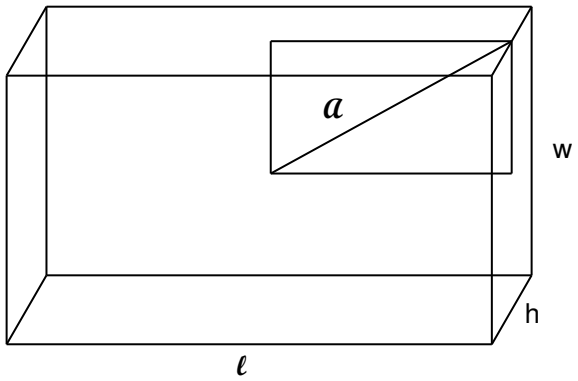


FIGURE A-3. Equipment dimensions.

Equations 5, 6, and 7 can be used to determine the minimum diameter of a coil for performance testing. The procedure is illustrated by the following example:

Given: Equipment dimensions are 1.4 x 0.9 x 0.7 meters and the coil is to produce a field of 1,600 A/m at the center and less than some larger predetermined value (4,000 A/m in this example) at the surface of the equipment nearest the coil.

Step 1 – Calculate "a" using equation 7:

$$a = \sqrt{\left(\frac{1.4}{2}\right)^2 + \left(\frac{0.9}{2}\right)^2}$$

$$= 0.83 \text{ m}$$

Step 2 – Determine NI from equation 6 when H = 1,600 A/m at the coil center:

$$NI = 2\,RH = 3{,}200\,R$$

Step 3 – Determine minimum R for coil by substituting NI and "a" into equation 5:

$$4{,}000 = \frac{3{,}200R}{4R}\left[\frac{2R^2 - (0.83)^2}{R^2 - (0.83)^2}\right]$$

$$4{,}000 = \left[\frac{2R^2 - 0.6925}{R^2 - 0.6925}\right]$$

Solving for R gives:

$$R = 0.94 \text{ m}$$

$$D = 2R = 1.88 \text{ m}$$

The coil diameter must be greater than 1.88 meters to produce a field of 1,600 A/m at the center and less than 4,000 A/m at the distance of 0.83 meters from the center. Maximum coil diameter is limited by available power or materials (cost).

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH

Preparing activity:
Navy – SH
(Project 1990-2022-003)

Review activities:
Army – MI
Navy – AS, EC, OS
DISA – DC5

Civil Agency:
GSA – FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:19Z
Check the source to verify that this is the current version before use.